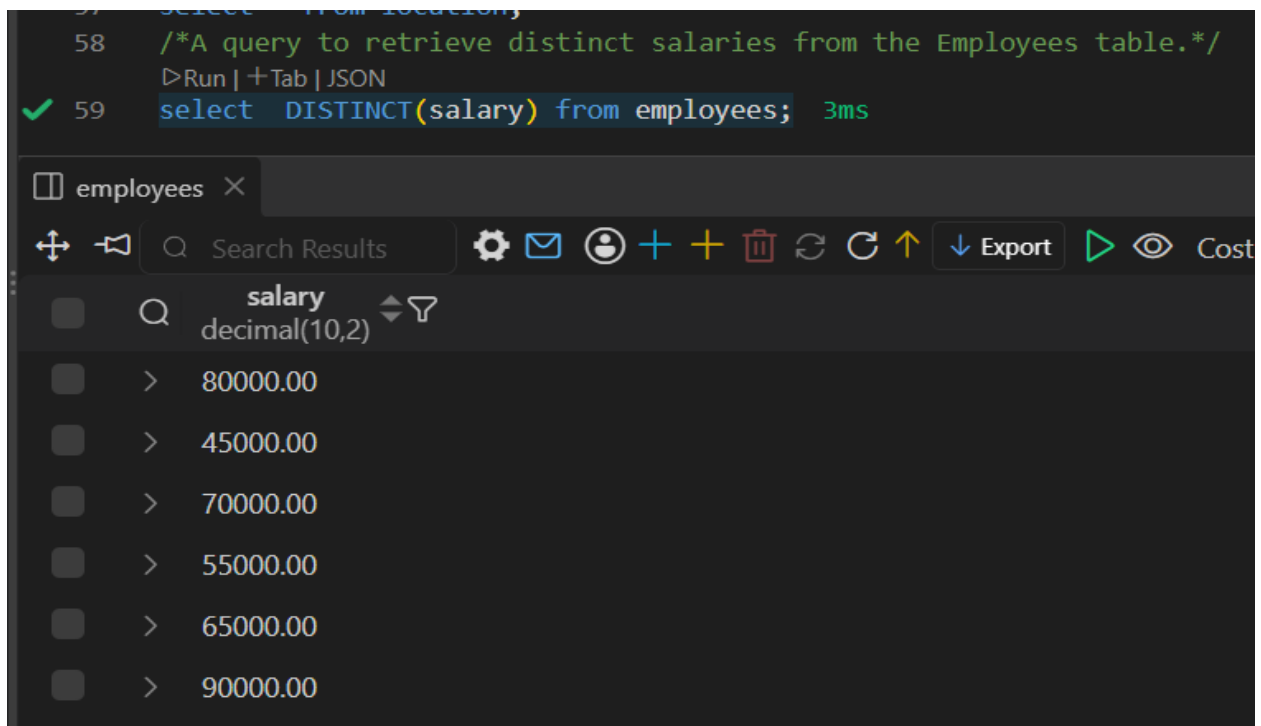


1.A query to retrieve distinct salaries from the Employees table.

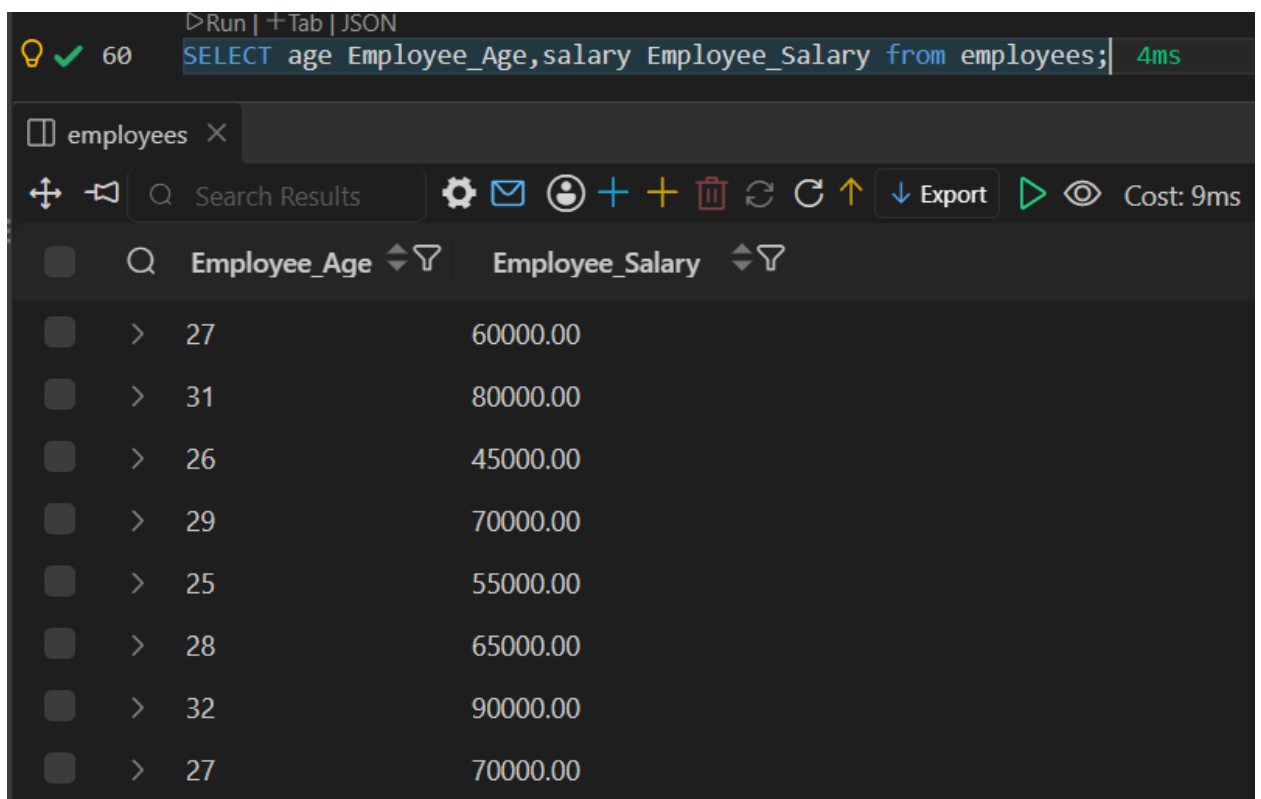
```
58 /*A query to retrieve distinct salaries from the Employees table.*/
    >Run | +Tab | JSON
✓ 59 select DISTINCT(salary) from employees; 3ms
```



salary
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00

2. Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
>Run | +Tab | JSON
💡 ✓ 60 SELECT age Employee_Age,salary Employee_Salary from employees; 4ms
```



Employee_Age	Employee_Salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00

3. Where Clause & Operators:

☉ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```
61 select * from employees
62 where salary > 50000 and hire_date < "2016-01-01"; 4ms
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5004	Kiara Malhotra	F	29	2015-07-05	(NULL)	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Exec	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00

☉ Find the employee whose designation is missing and fill it with "Data Scientist".

```
65 update employees
66 set designation = "Data Scientist"
67 where designation is null; 3ms
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	10	2	45000.00
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Exec	11	1	55000.00

4.ORDER BY: Find employees sorted by department ID in ascending order and salary in descending order.

```
68 SELECT department_id,salary from employees
69 ORDER BY department_id asc,salary desc; 4ms
```

department_id	salary
1	90000.00
2	90000.00
2	70000.00
2	60000.00
3	100000.00
3	75000.00

5. Display the first 5 employees hired in the year 2018.

```
Run | +Tab | JSON
70 select * from employees
71 where YEAR(hire_date) = "2018"
72 LIMIT 5; 3ms
```

	employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
>	5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
>	5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
>	5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
>	5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
>	5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00

6. Calculate the sum of all salaries in the Finance department.

```
Run | +Tab | JSON
73 select sum(salary) as Total_Salary from employees
74 where department_id = (select department_id from departments where department_name = "Finance"); 4ms
```

Total_Salary
170000.00

7. Find the minimum age among all employees.

```
Run | +Tab | JSON
75 select min(age) as minium_age from employees; 5ms
76
```

minium_age
25

8. List the maximum salary for each location.

```
Run | +Tab | JSON
76 SELECT location_id,max(salary) as Max_salary from employees
77 group by location_id; 4ms
```

location_id	Max_salary
1	95000.00
2	95000.00
3	90000.00
4	100000.00

9. Calculate the average salary for each designation containing the word 'Analyst'.

```
Run | +Tab | JSON
78 select designation,AVG(salary) Avg_salary from employees
79 where designation like "%Analyst"
80 GROUP BY designation; 5ms
```

designation	Avg_salary
Business Analyst	75000.000000
Data Analyst	60000.000000
Finance Analyst	85000.000000
Financial Analyst	85000.000000
Marketing Analyst	60000.000000
Quality Assurance Analyst	80000.000000
Supply Chain Analyst	60000.000000

10. Find departments with less than 3 employees.

```
Run | +Tab | JSON
81 select department_id,count(employee_id) as emp_cnt from employees
82 GROUP BY department_id
83 HAVING emp_cnt < 3; 3ms
```

department_id	emp_cnt
1	1
5	2
6	2
7	2
9	1
11	2
13	1

11. Find locations with female employees whose average age is below 30.

```

84 select location_id, avg(age) avg_age from employees
85 where gender = "F"
86 group by location_id
87 having avg_age < 30;

```

location_id	avg_age
1	26.7500
2	27.3333
4	29.2000

12. List employee names, their designations, and department names where employees are assigned to a department.

```

88 /*List employee names, their designations, and department names where employees are assigned to a department.*/
89 select e.employee_name, e.designation, d.department_name from employees e
90 join departments d on e.department_id = d.department_id;
91

```

employee_name	designation	department_name
Anvi Chaudhary	Business Development Exec	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science

13. List all departments along with the total number of employees in each department, including departments with no employees.

```
90  /*List all departments along with the total number of employees in each department, including departments with no employees.*/
    ▷Run|+Tab|JSON
91  select d.department_name,count(e.employee_id) emp_cnt from departments d
92  left join employees e on e.department_id = d.department_id
93  GROUP by d.department_name;
94
```

Result(RO) X

Search Results

Cost: 2ms < 1 > Total 13

Export

department_name	emp_cnt
varchar	bigint
> Business Development	2
> Content Creation	2
> Customer Support	3
> Data Science	3
> Design	3
> Finance	2
> Human Resources	3
> IT	4
> Marketing	3
> Operations	1

14. Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
94  /*14. Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.*/
    ▷Run|+Tab|JSON
95  select l.location,e.employee_name from employees e
96  right join location l on e.location_id = l.location_id ; 1ms
97
```

Result(RO) X

Search Results

Cost: 1ms < 1 > Total 30

Export

location	employee_name
varchar	varchar
> Bangalore	Aaradhya Iyer
> Bangalore	Dhruv Shetty
> Bangalore	Myra Verma
> Bangalore	Vihaan Mohan
> Bangalore	Kabir Nair
> Bangalore	Anika Paul
> Bangalore	Vivaan Singh
> Bangalore	Arjun Kumar
> Chennai	Reyansh Singh
> Chennai	Anvi Chaudhary